

HAT-P-36b

R-0896

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_240208 · created 2026-07-29T02:50:39+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460348.9458 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1566 +/- 0.0062
Transit depth (Rp/Rs)^2	2.45 +/- 0.19 [%]
Orbital Inclination (inc)	86.04 +/- 1.66
Airmass coefficient 1 (a1)	1.095 +/- 0.011
Airmass coefficient 2 (a2)	-0.082 +/- 0.036
Scatter in the residuals of the lightcurve fit is	0.99 %
Best Comparison Star	#2 - [111, 203]
Optimal Aperture	4.45
Optimal Annulus	6.6
Transit Duration (day)	0.0966 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1566 $\pm$ 0.0062 vs 0.126 (deviation 4.43 σ)
Duration fit vs published	0.0966 d vs 0.0925 d (deviation 1.12 σ)
Mid-time O-C	8.5 min
Depth SNR	22.7
Residual scatter	0.99 %

Light curve

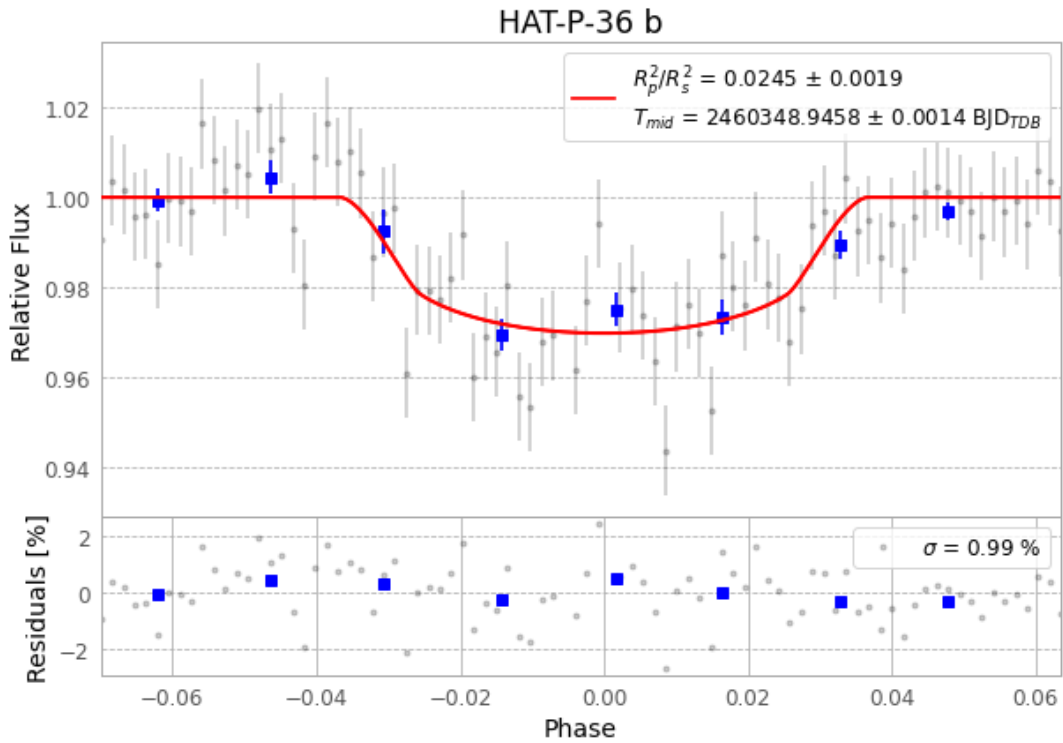


Figure 1 — FinalLightCurve_HAT-P-36 b_2024-02-08.png (SHA-256 f1c703021963f373...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [383, 266], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3096e23a55906e1f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

86 FITS frames (56 MB), HATP-36240208082116.FITS → HATP-36240208123615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-240208.log (2669034e0bb9...), FinalLightCurve_HAT-P-36 b_2024-02-08.png (f1c703021963...), FinalLightCurve_HAT-P-36 b_2024-02-08.csv (a70d3ae7ed9f...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 02:50Z.